the positive electrode is pulled away against its spring pressure through a small distance by coil and thus an arc is struck between electrodes. This arc is maintained by transfer of carbon particles from one electrode to other electrode. These particles travel from positive electrode to negative electrode, thus after sometime of operation positive electrode becomes hollow and negative becomes pointed. That's why positive electrode is made double than negative electrode. In carbon arc lamp 85 % of light is given by positive electrode which produces high intensity light and only 10% by negative electrode and 5% by air. The temperature of the positive electrode is 4000°C and that of the negative electrode is about 2500°C. The luminous efficiency of such lamps is approximately 9 lumen / watt.

2.5 Incandescent lamp

The filament of this lamp is heated up to the incandescent stage of heat. So these types of lamps are called as incandescent lamps. There are two types.

1. Vacuum type lamp and
2. Gas filled type lamp

In this type of lamp, the sphere shaped glass cover is used. The glass stem is fixed in the centre of the lamp. This stem supports wires in holding the filament. The top of the lamp is sealed. Pins are used for holding the lamp in the holder.

2.5.1 Vacuum lamp

In this lamp, the air is evacuated to protect the filament from burning by oxygen mixed in the air. Vacuum lamp is shown in figure 2.4 and 2.4(a).



Fig. 2.4 Vacuum lamp

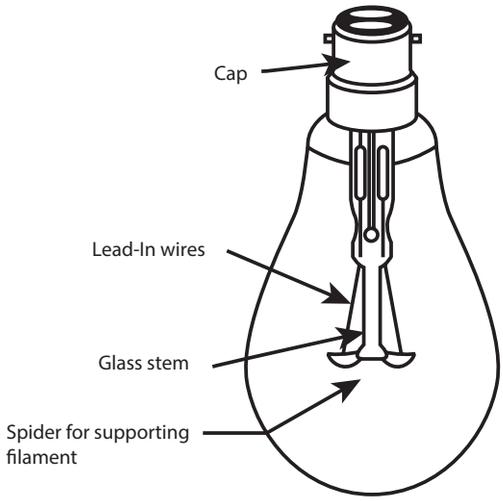


Fig. 2.4(a) Line diagram of vacuum lamp

When the filament is heated, due to the current that passes through the high resistance filament, the moving electron creates friction. So the heat is generated in the filament. When the heat raises up to the incandescent stage, the light is emitted from the filament. The emitted light is reflected by a sphere-shaped glass cover.

2.5.2 Gas filled lamp

It is shown in figure 2.5. In the evacuated lamp, the filament evaporates and deposits inside of the glass cover after long use and makes black shade on the glass

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